BPM1
CPR IMCQ 8
Physiology questions
(13 Questions)
KEY

1. A 65-year-old woman comes to the physician because of a 5-week history of shortness of breath and swollen legs. She has a 6-year history of hypertension and is not adherent to medications. Her pulse is 92/min, respirations are 20/min, and blood pressure is 140/98 mm Hg. Physical examination shows decreased breath sounds at the lung bases bilaterally and pitting edema of the legs. She is diagnosed with pulmonary edema due to heart failure. Which of the following changes in the lungs best explains the genesis of this patient's condition?

A. Increased capillary permeability
B. Increased capillary hydrostatic pressure
C. Decreased capillary oncotic pressure
D. Decreased lymphatic drainage
E. Increased lung surface tension

1. A 65-year-old woman comes to the physician because of a 5-week history of shortness of breath and swollen legs. She has a 6-year history of hypertension and is not adherent to medications. Her pulse is 92/min, respirations are 20/min, and blood pressure is 140/98 mm Hg. Physical examination shows decreased breath sounds at the lung bases bilaterally and pitting edema of the legs. She is diagnosed with pulmonary edema due to heart failure. Which of the following changes in the lungs best explains the genesis of this patient's condition?

A. Increased capillary permeability
✓ B. Increased capillary hydrostatic pressure
C. Decreased capillary oncotic pressure
D. Decreased lymphatic drainage
E. Increased lung surface tension

SOM.MK.II.BPM1.3.CPR.2.PHYS.0218. Explain the development of pulmonary edema by a) increased hydrostatic pressure, b) increased permeability, c) impaired lymphatic outflow or increased central venous pressure, and d) hemodilution (e.g., with saline volume resuscitation).

2. A 24-year-old woman is brought to the emergency department by her roommate because of a 1-hour history of difficulty breathing and altered mental status. Her roommate says that she found an empty container of sleeping pills near her bed. Her respirations are 8/min with a reduced tidal volume, blood pressure is 126/82 mm Hg, and pulse is 92/min. Physical examination shows hypoventilation and reduced response to verbal and tactile stimuli. She is diagnosed with respiratory depression due to an overdose of sleeping pills. Which of the following arterial blood gas alterations is most likely in this patient?

Option	PaO_2	$PaCO_2$	pH
A	↓	↓	↓
B	↓	↑	↑
C	↑	↓	↓
D	↑	↑	↓
E	↓	↑	↓

2. A 24-year-old woman is brought to the emergency department by her roommate because of a 1-hour history of difficulty breathing and altered mental status. Her roommate says that she found an empty container of sleeping pills near her bed. Her respirations are 8/min with a reduced tidal volume, blood pressure is 126/82 mm Hg, and pulse is 92/min. Physical examination shows hypoventilation and reduced response to verbal and tactile stimuli. She is diagnosed with respiratory depression due to an overdose of sleeping pills. Which of the following arterial blood gas alterations is most likely in this patient?

Option	PaO2	PaCO2	pH
A	↓	↓	↓
B	↓	↑	↑
C	↑	↓	↓
D	↑	↑	↓
✔ E	↓	↑	↓

SOM.MK.II.BPM1.3.CPR.2.PHYS.0349. Describe how hypoventilation would affect alveolar and arterial PO2 and PCO2 and the A-a gradient.

3. A 15-year-old girl is brought to the physician because of a 4-month history of shortness of breath on exertion. Menarche occurred 2 years ago, and her menses last for 6-8 days with heavy blood loss. Her pulse is 104/min and respirations are 24/min; other vital signs are within normal limits. Physical examination shows pale tongue, nails and conjunctiva. Laboratory studies show a decreased hemoglobin concentration and low hematocrit. A peripheral blood smear shows iron deficiency anemia. Arterial blood gas analysis on room air shows a normal PaO2 but a decreased PaCO2. This altered gas in the blood is transported mostly in which of the following forms?

A. Carbamino compound
B. Carbonate
C. Carbonic acid
D. Bicarbonate
E. Dissolved in plasma

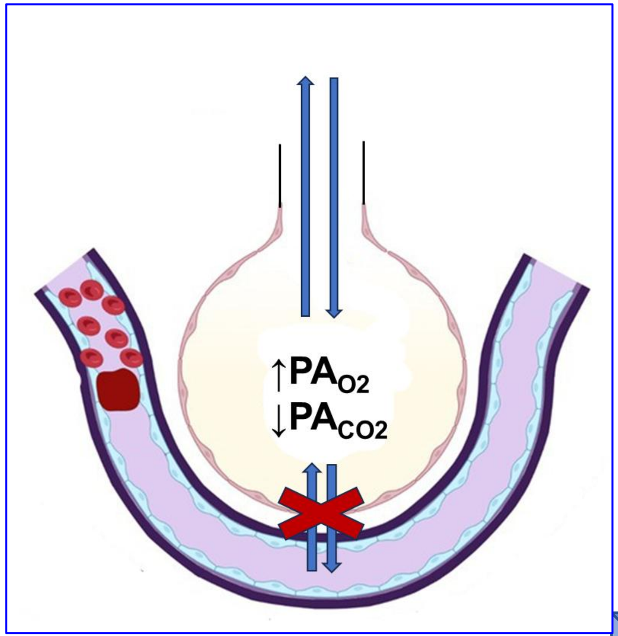
3. A 15-year-old girl is brought to the physician because of a 4-month history of shortness of breath on exertion. Menarche occurred 2 years ago, and her menses last for 6-8 days with heavy blood loss. Her pulse is 104/min and respirations are 24/min; other vital signs are within normal limits. Physical examination shows pale tongue, nails and conjunctiva. Laboratory studies show a decreased hemoglobin concentration and low hematocrit. A peripheral blood smear shows iron deficiency anemia. Arterial blood gas analysis on room air shows a normal PaO2 but a decreased PaCO2. This altered gas in the blood is transported mostly in which of the following forms?

A. Carbamino compound
B. Carbonate
C. Carbonic acid
✓ D. Bicarbonate
E. Dissolved in plasma

SOM.MK.II.BPM1.3.CPR.2.PHYS.0266. List the forms in which carbon dioxide is carried in the blood. Identify the percentage of total CO2 transported as each form

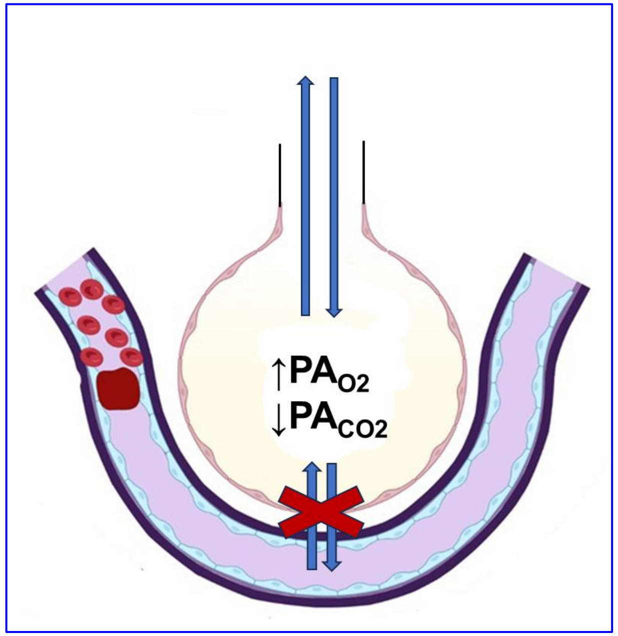
4. A 38-year-old woman is brought to the emergency department because of a 6-hour history of shortness of breath. She has a 2-year history of deep vein thrombosis and takes blood thinner. Her blood pressure is 124/84 mmHg, pulse is 105/min, and respirations are 22/min. A CT scan with pulmonary angiography shows blocks in the segmental branches of the left pulmonary artery, partially reducing the blood flow to a group of alveoli in the left lower lung. Which of the following sets of partial pressures of O_2 and CO_2 in mm Hg is most likely in the affected alveoli of the left lung in this patient? (Normal PA_{O2}=100; Normal PA_{CO2}=40)

	Option	Pco_2	Po_2	
A.	A	0	50	20%
B.	B	07	28	20%
C.	C	40	100	20%
D.	D	08	122	20%
E.	E	70	150	20%



4. A 38-year-old woman is brought to the emergency department because of a 6-hour history of shortness of breath. She has a 2-year history of deep vein thrombosis and takes blood thinner. Her blood pressure is 124/84 mmHg, pulse is 105/min, and respirations are 22/min. A CT scan with pulmonary angiography shows blocks in the segmental branches of the left pulmonary artery, partially reducing the blood flow to a group of alveoli in the left lower lung. Which of the following sets of partial pressures of O_2 and CO_2 in mm Hg is most likely in the affected alveoli of the left lung in this patient? (Normal PA_{O2}=100; Normal PA_{CO2}=40)

Option	Pco_2	Po_2
A	0	50
B	07	28
C	40	100
D ✓	08	122
E	70	150



SOM.MK.II.BPM1.3.CPR.2.PHYS.0246. Define the ventilation/perfusion (V/Q) ratio of an alveolar-capillary lung unit and show how it determines the PO2 and PCO2 of the blood emerging from that lung unit.

5. A 36-year-old healthy man comes to the physician for a routine examination prior to participating in a 15K marathon for the first time. His pulse is 60/min, respirations are 14/min. A pulse oximetry on room air shows an O_2 saturation of 98%. He asks if 100% O_2 would enhance his performance after seeing an advertisement about an oxygen bar. Which of the following outcomes is most likely to occur due to this therapy?

A. Increased O2 carried by Hb
B. Decreased dissolved O2
C. No appreciable increase in O2 content
D. A 5-fold increase in O2 content
E. An appreciable change in the PaO2

5. A 36-year-old healthy man comes to the physician for a routine examination prior to participating in a 15K marathon for the first time. His pulse is 60/min, respirations are 14/min. A pulse oximetry on room air shows an O_2 saturation of 98%. He asks if 100% O_2 would enhance his performance after seeing an advertisement about an oxygen bar. Which of the following outcomes is most likely to occur due to this therapy?

A. Increased O2 carried by Hb
B. Decreased dissolved O2
✓ C. No appreciable increase in O2 content
D. A 5-fold increase in O2 content
E. An appreciable change in the PaO2

SOM.MK.II.BPM1.3.CPR.2PHYS.0256. Draw an oxyhemoglobin dissociation curve (hemoglobin oxygen equilibrium curve) showing the relationships between oxygen partial pressure, hemoglobin saturation, and blood oxygen content.

6. A 27-year-old woman comes to the emergency department because of a 10-hour history of headaches and shortness of breath. Two days ago, she came from the sea level to a place which is at an altitude of 12,000 feet. Her respirations are 26/min with an increased tidal volume. Physical examination shows hyperventilation. Which of the following alterations is most likely in this patient?

A. Increased $PaCO_2$
B. Vasodilation in the pulmonary blood vessels
C. Increased A-a gradient
D. Increased blood pH
E. Reduced cardiac output

6. A 27-year-old woman comes to the emergency department because of a 10-hour history of headaches and shortness of breath. Two days ago, she came from the sea level to a place which is at an altitude of 12,000 feet. Her respirations are 26/min with an increased tidal volume. Physical examination shows hyperventilation. Which of the following alterations is most likely in this patient?

A. Increased $PaCO_2$
B. Vasodilation in the pulmonary blood vessels
C. Increased A-a gradient
✓ D. Increased blood pH
E. Reduced cardiac output

SOM.MK.II.BPM1.3.CPR.2.PHYS.0254. Describe the mechanisms for the shift in alveolar ventilation that occur immediately upon ascent to high altitude, after remaining at altitude for two weeks, and immediately upon return to sea level

7. A 4-year-old boy is brought to the emergency department by his mother because of a 30-minute history of severe difficulty breathing. The mother says that he was playing with plastic beads and accidentally swallowed one of them. His respirations are 40/min with increased tidal volume and an oxygen saturation of 89% on room air. Physical examination shows diminished breath sounds in the right upper lung fields. Imaging shows plastic bead being stuck in the right upper lobar bronchus. Which of the following sets of alterations in partial pressures of gases is most likely in the affected alveoli and systemic blood in this patient?

Option	Alveolar PO2	Alveolar PCO2	Arterial PO2
A	↓	↓	↓
B	↓	↑	↓
C	↑	↑	↑
D	↓	↑	↑
E	No change	↑	↓

7. A 4-year-old boy is brought to the emergency department by his mother because of a 30-minute history of severe difficulty breathing. The mother says that he was playing with plastic beads and accidentally swallowed one of them. His respirations are 40/min with increased tidal volume and an oxygen saturation of 89% on room air. Physical examination shows diminished breath sounds in the right upper lung fields. Imaging shows plastic bead being stuck in the right upper lobar bronchus. Which of the following sets of alterations in partial pressures of gases is most likely in the affected alveoli and systemic blood in this patient?

Option	Alveolar PO2	Alveolar PCO2	Arterial PO2
A	↓	↓	↓
B ✔	↓	↑	↓
C	↑	↑	↑
D	↓	↑	↑
E	No change	↑	↓

SOM.MK.II.BPM1.3.CPR.2.PHYS.0236. Explain the relationship between alveolar ventilation and the arterial PCO2 and PO2 and pH, compliance and ventilation. 2.Describe how hypoventilation would affect alveolar and arterial PO2 and PCO2 and the A-a gradient.

8. A 79-year-old man comes to the physician because of a 4-week history of passing blood in stools, dizziness, and shortness of breath on exertion. His blood pressure is 130/76 mm Hg, pulse is 86/min, and respirations are 22/min. Physical examination shows pallor and a palpable mass in the right lower quadrant of the abdomen. Laboratory studies show decreased hematocrit and red cell count but increased erythrocyte sedimentation rate and serum erythropoietin. Colonoscopy shows a mass in the sigmoid colon. Examination of a biopsy specimen of this mass shows adenocarcinoma. Which of the following additional laboratory alterations related to anemia is most likely in this patient?

A. Increased O_2 carrying capacity of blood
B. Normal percentage O_2 saturation of hemoglobin
C. Decreased PaO_2
D. Increased blood viscosity
E. Increased O_2 content of blood

8. A 79-year-old man comes to the physician because of a 4-week history of passing blood in stools, dizziness, and shortness of breath on exertion. His blood pressure is 130/76 mm Hg, pulse is 86/min, and respirations are 22/min. Physical examination shows pallor and a palpable mass in the right lower quadrant of the abdomen. Laboratory studies show decreased hematocrit and red cell count but increased erythrocyte sedimentation rate and serum erythropoietin. Colonoscopy shows a mass in the sigmoid colon. Examination of a biopsy specimen of this mass shows adenocarcinoma. Which of the following additional laboratory alterations related to anemia is most likely in this patient?

A. Increased O_2 carrying capacity of blood
B. Normal percentage O_2 saturation of hemoglobin ✓
C. Decreased PaO_2
D. Increased blood viscosity
E. Increased O_2 content of blood

SOM.MK.II.BPM1.3.CPR.2.PHYS.0255. Define oxygen partial pressure (tension), oxygen content, and percent hemoglobin saturation as they pertain to blood.

9. A 26-year-old healthy man participates in a study that is related to the pulmonary circulation. His vital signs are within normal limits. He is asked to breath forcefully and the mean pulmonary arterial pressure, heart rate, and cardiac output are recorded. Which of the following sets of alterations in the pulmonary vessels and pulmonary vascular resistance is most likely when this participant forcefully exhales and reaches residual volume?

Option	Alveolar Vessel Status	Extra-alveolar Vessel Status	Pulmonary vascular resistance
A	**Distended**	**Compressed**	↑
B	**Compressed**	**Compressed**	↓
C	**Distended**	**Distended**	↓
D	**Compressed**	**Distended**	↑
E	**Distended**	**No change**	**No change**
F	**Compressed**	**Compressed**	↑

9. A 26-year-old healthy man participates in a study that is related to the pulmonary circulation. His vital signs are within normal limits. He is asked to breath forcefully and the mean pulmonary arterial pressure, heart rate, and cardiac output are recorded. Which of the following sets of alterations in the pulmonary vessels and pulmonary vascular resistance is most likely when this participant forcefully exhales and reaches residual volume?

Option	Alveolar Vessel Status	Extra-alveolar Vessel Status	Pulmonary vascular resistance
✓ A	**Distended**	**Compressed**	↑
B	**Compressed**	**Compressed**	↓
C	**Distended**	**Distended**	↓
D	**Compressed**	**Distended**	↑
E	**Distended**	**No change**	**No change**
F	**Compressed**	**Compressed**	↑

SOM.MK.II.BP M1.3.CPR.2.PHYS.0216. Explain how pulmonary vascular resistance in alveolar and extra-alveolar blood vessels changes with lung volume.

10. A 67-year-old man is brought to the emergency department because of a 2-week history of productive cough and worsening shortness of breath. He has a 25-year history of COPD. Physical examination shows labored breathing and inspiratory crackles in both lung fields. Results of arterial blood gas analysis on room air when the patient arrived and repeated one hour after he is started on 30% supplemental oxygen by nasal cannula are shown. Partial pressure of inspired air (P_{IO2}) on room air is 214 mm Hg and Respiratory quotient (R) is 0.8. Which of the following values best represents this patient's PA_{O2} after treatment with the supplemental oxygen?

A. 50 mmHg
B. 60 mmHg
C. 80 mmHg
D. 164 mmHg
E. 214 mmHg

ABG results	**Without supplemental O2**	**With 30% supplemental O2**
PaO2 (mmHg)	61	95
PaCO2 (mmHg)	64	40
Arterial HCO3 (mmol/L)	26	24
O2 Saturation (%)	78	91

Alveolar Gas Equation:
$PA_{O2} = P_{IO2} - Pa_{CO2}/R$

10. A 67-year-old man is brought to the emergency department because of a 2-week history of productive cough and worsening shortness of breath. He has a 25-year history of COPD. Physical examination shows labored breathing and inspiratory crackles in both lung fields. Results of arterial blood gas analysis on room air when the patient arrived and repeated one hour after he is started on 30% supplemental oxygen by nasal cannula are shown. Partial pressure of inspired air (P_{IO2}) on room air is 214 mm Hg and Respiratory quotient (R) is 0.8. Which of the following values best represents this patient's PA_{O2} after treatment with the supplemental oxygen?

ABG results	Without supplemental O2	With 30% supplemental O2
PaO2 (mmHg)	61	95
PaCO2 (mmHg)	64	40
Arterial HCO3 (mmol/L)	26	24
O2 Saturation (%)	78	91

A. 50 mmHg
B. 60 mmHg
C. 80 mmHg
D. 164 mmHg ✔
E. 214 mmHg

Alveolar Gas Equation:
$PA_{O2} = P_{IO2} - Pa_{CO2}/R$

SOM.MK.II.BPM1.3.CPR.2.PHYS.0238.
Calculate the alveolar oxygen partial pressure (PAO2) and the amount of supplemental O2 required to overcome a reduction in PAO2 caused by hypoventilation or high altitude using the simplified form of the alveolar gas equation.

11. A 68-year-old woman comes to the physician because of a 6-month history of increasing shortness of breath, fatigue, and a dry cough. She works in a factory that manufactures asbestos sheets. Her blood pressure is 136/84 mmHg, respirations are 22/min and shallow, and pulse is 86/min. Physical examination shows finger clubbing and diffuse lung crackles bilaterally. Diffusing capacity of lung (DL_{CO}) is performed. During this test, she is made to breathe a mixture of gases containing carbon monoxide (CO) and is asked to hold her breath for 10 seconds. Her P_ACO is 0.4 mmHg, and the CO uptake is 12 mL/min. Consider PaCO = Zero (0). Which of the following values best approximates DL_{CO} of this patient?

A. 10 mL/min

B. 20 mL/min

C. 30 mL/min

D. 40 mL/min

E. 50 mL/min

V_{gas} = DL X [P1 – P2] or DL = V_{gas}/ [P1 – P2]

DL = 12/0.4 = 30 mL/min/mm Hg

11. A 68-year-old woman comes to the physician because of a 6-month history of increasing shortness of breath, fatigue, and a dry cough. She works in a factory that manufactures asbestos sheets. Her blood pressure is 136/84 mmHg, respirations are 22/min and shallow, and pulse is 86/min. Physical examination shows finger clubbing and diffuse lung crackles bilaterally. Diffusing capacity of lung (DL_{CO}) is performed. During this test, she is made to breathe a mixture of gases containing carbon monoxide (CO) and is asked to hold her breath for 10 seconds. Her P_ACO is 0.4 mmHg, and the CO uptake is 12 mL/min. Consider PaCO = Zero (0). Which of the following values best approximates DL_{CO} of this patient?

A. 10 mL/min
B. 20 mL/min
✓C. 30 mL/min
D. 40 mL/min
E. 50 mL/min

SOM.MK.II.BPM1.3.CPR.2.PHYS.0265. Calculate the diffusing capacity of carbon monoxide (DLCO) from the alveolar to arterial PO2 difference. Describe the normal value for DLCO and the significance of an elevated DO2.

V_{gas} = DL X [P1 – P2] or DL = V_{gas}/ [P1 – P2]

DL = 12/0.4 = 30 mL/min/mm Hg

12. A 61-year-old man is brought to the emergency department by his wife because of a 2-hour history of shortness of breath and altered mental status. He is an intravenous drug user. His blood pressure is 132/84, pulse is 96/min and respirations are 9/min with a reduced tidal volume. Physical examination shows bluish coloration of the tongue, conjunctiva, and nailbeds. Laboratory studies show a PaO_2 of 72 mmHg (Normal:85-95 mm Hg), $PaCO_2$ of 66 mm Hg (Normal: 40-44), and an A-a gradient of 8 mm Hg (Normal: 5-10 mm Hg). Which of the following mechanisms best explains the laboratory findings in this patient?

A. Diffusion defect in the lung
B. Generalized hypoventilation
C. Regional increase in V/Q ratio
D. Intrapulmonary shunt
E. Fluid in lung interstitial space

12. A 61-year-old man is brought to the emergency department by his wife because of a 2-hour history of shortness of breath and altered mental status. He is an intravenous drug user. His blood pressure is 132/84, pulse is 96/min and respirations are 9/min with a reduced tidal volume. Physical examination shows bluish coloration of the tongue, conjunctiva, and nailbeds. Laboratory studies show a PaO2 of 72 mmHg (Normal:85-95 mm Hg), PaCO2 of 66 mm Hg (Normal: 40-44), and an A-a gradient of 8 mm Hg (Normal: 5-10 mm Hg). Which of the following mechanisms best explains the laboratory findings in this patient?

A. Diffusion defect in the lung
✓ B. Generalized hypoventilation
C. Regional increase in V/Q ratio
D. Intrapulmonary shunt
E. Fluid in lung interstitial space

SOM.MK.II.BPM1.3.CPR.2.PHYS.0252. Calculate the alveolar to arterial PO2 difference, (A-a) O2 gradient. List the normal value range for (A-a) O2 and explain the significance of an elevated (A-a) O2.

13. A 53-year-old man is brought to the emergency department because of a 2-day history of cough and progressive shortness of breath. He has a 7-day history of fever, runny nose, muscle aches, and photophobia. His pulse is 106/min, blood pressure is 142/78 mm Hg, respirations are 26/min with shallow tidal volume, and an oxygen saturation of 80% on room air. Physical examination shows central cyanosis, respiratory distress with use of accessory muscles of breathing, and altered mental status. Chest x-ray shows diffuse opacities bilaterally. Pulmonary studies show decreased DL_{CO} and P_{aO2}, but increased A-a gradient. He is diagnosed with acute respiratory distress syndrome (ARDS). He is started on 40% supplemental oxygen by nasal cannula. His P_{aO2} and A-a gradient do not improve and he dies. Which of the following defects is the most likely cause of this patient's hypoxemia?

A. Pulmonary fibrosis
B. Generalized hypoventilation
C. V/Q mismatch
D. Diffusion defect
E. Pulmonary shunt

13. A 53-year-old man is brought to the emergency department because of a 2-day history of cough and progressive shortness of breath. He has a 7-day history of fever, runny nose, muscle aches, and photophobia. His pulse is 106/min, blood pressure is 142/78 mm Hg, respirations are 26/min with shallow tidal volume, and an oxygen saturation of 80% on room air. Physical examination shows central cyanosis, respiratory distress with use of accessory muscles of breathing, and altered mental status. Chest x-ray shows diffuse opacities bilaterally. Pulmonary studies show decreased DL_{CO} and P_{aO2}, but increased A-a gradient. He is diagnosed with acute respiratory distress syndrome (ARDS). He is started on 40% supplemental oxygen by nasal cannula. His P_{aO2} and A-a gradient do not improve and he dies. Which of the following defects is the most likely cause of this patient's hypoxemia?

A. Pulmonary fibrosis
B. Generalized hypoventilation
C. V/Q mismatch
D. Diffusion defect
✓ E. Pulmonary shunt

SOM.MK.II.BPM1.3.CPR.2.PHYS.0253. Describe how changes to the A-a gradient occur and what factors lead to an increased A-a gradient in patients.

CPR IMCQ 8
Fall 2025

Biochemistry 3 Questions

KEY file

14. A 25-year-old woman is brought to the emergency room because of a 3-hour history of headaches, weakness and shortness of breath. She has a 10-year history of bronchial asthma. Vital signs: Pulse: 105/min, Respirations are 20/min, Temperature: 37ºC (98.6ºF). Pulse oximetry on room air shows an oxygen saturation of 85%. Physical examination shows bilateral lung wheezing and increased expiration time when breathing. Laboratory studies show:
Fasting glucose: 100 mg/dl (70-100 mg/dl)
Serum potassium (K^+): 5.4 mEq/L (3.5-5.5 mEq/L)
Serum lactate: Elevated
Activity of which of the following enzymes under anaerobic conditions is dependent on the formation of lactate?

A. Pyruvate kinase
B. Glyceraldehyde 3-phosphate dehydrogenase
C. Phosphofructokinase-1
D. Glucose 6-phosphate dehydrogenase
E. Phosphoglycerate kinase

14. A 25-year-old woman is brought to the emergency room because of a 3-hour history of headaches, weakness and shortness of breath. She has a 10-year history of bronchial asthma. Vital signs: Pulse: 105/min, Respirations are 20/min, Temperature: 37ºC (98.6ºF). Pulse oximetry on room air shows an oxygen saturation of 85%. Physical examination shows bilateral lung wheezing and increased expiration time when breathing. Laboratory studies show:

Fasting glucose: 100 mg/dl (70-100 mg/dl)

Serum potassium (K+): 5.4 mEq/L (3.5-5.5 mEq/L)

Serum lactate: Elevated

Activity of which of the following enzymes under anaerobic conditions is dependent on the formation of lactate?

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0301; Explain significance of lactate production in anaerobic glycolysis and describe the fate of lactate formed in RBC and muscle (Cori cycle)
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0308; Identify the biochemical mechanisms involved in the occurrence of lactic acidosis in states of poor tissue perfusion/ oxygenation.

A. Pyruvate kinase
B. ✓ Glyceraldehyde 3-phosphate dehydrogenase
C. Phosphofructokinase-1
D. Glucose 6-phosphate dehydrogenase
E. Phosphoglycerate kinase

15. An 8-year-old girl is brought to the physician by her parents because of a 2-month history of tiredness and pale appearance. She is unable to participate in school sports. Physical examination shows conjunctival pallor and mild scleral icterus. Laboratory studies show elevated serum LDH levels and increased erythrocyte 2,3-bisphosphoglycerate (2,3-BPG) levels. Which of the following is most likely observed in the erythrocytes of this patient?

A. Accumulation of oxidants
B. Accumulation of HbH
C. Beta globin tetramers
D. Reduced ATP production
E. Polymerization of hemoglobin
F. Heinz bodies

15. An 8-year-old girl is brought to the physician by her parents because of a 2-month history of tiredness and pale appearance. She is unable to participate in school sports. Physical examination shows conjunctival pallor and mild scleral icterus. Laboratory studies show elevated serum LDH levels and increased erythrocyte 2,3-bisphosphoglycerate (2,3-BPG) levels. Which of the following is most likely observed in the erythrocytes of this patient?

SOM.1a.BPM1.3.CPR.2.BCHM. RS. 1513 Differentiate between pyruvate kinase deficiency (defect in glycolytic pathway) and glucose 6-phosphate dehydrogenase deficiency (pentose phosphate defect)

A. Accumulation of oxidants
B. Accumulation of HbH
C. Beta globin tetramers
D. Reduced ATP production
E. Polymerization of hemoglobin
F. Heinz bodies

16.A 14-year-old boy is brought to the physician by his father for a follow-up examination. He is being investigated for mild jaundice and anemia. There is no history of major medical illness in the family. Physical examination shows mild yellowing of the eyes. Laboratory studies for activity of the enzymes involved in the glycolytic pathway shows reduced phosphoglycerate kinase activity. Which of the following nucleotide-derived molecules is most likely necessary for the biochemical reaction involving the defective enzyme?

A. NADH
B. AMP
C. $NADP^+$
D. ADP
E. NAD^+
F. NADPH

16. A 14-year-old boy is brought to the physician by his father for a follow-up examination. He is being investigated for mild jaundice and anemia. There is no history of major medical illness in the family. Physical examination shows mild yellowing of the eyes. Laboratory studies for activity of the enzymes involved in the glycolytic pathway shows reduced phosphoglycerate kinase activity. Which of the following nucleotide-derived molecules is most likely necessary for the biochemical reaction involving the defective enzyme?

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0300; Define substrate level phosphorylation and identify those reactions in glycolytic pathway

A. NADH
B. AMP
C. $NADP^+$
D. ADP
E. NAD^+
F. NADPH

KEY

1. B
2. E
3. D
4. D
5. C
6. D
7. B
8. B
9. A
10. D
11. C
12. B
13. E
14. B
15. D
16. D